



Adaptive open domain recognition by coarse-to-fine prototype-based network^{*}

Yuan Yuan, Xinxing He, Zhiyu Jiang^{*}

School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, PR China

ARTICLE INFO

Article history:

Received 3 September 2021

Revised 4 February 2022

Accepted 14 March 2022

Available online 16 March 2022

Keywords:

Open domain recognition

Image classification

Adaptive openness

Prototype learning

Unknown class recognition

ABSTRACT

Open domain recognition has attracted great attention in recent two years, which aims to assign a specific identification for each target sample in the presence of large domain discrepancy both in label space and data distributions. Most existing approaches rely on abundant prior information about the relationship of the label sets between the source and the target domain, which is a great limitation for their applications in practical wild. In this paper, a new Adaptive Open Domain Recognition (AODR) task is introduced, which can generalize to various openness and requires no prior information on the label set. To achieve this adaptive transfer task, a two-stage Progressive Adaptation Network is designed, whose learning process consists of multiple episodes. Each episode is performed to simulate an AODR task. Through training and refining multiple episodes, the basic model has progressively accumulated wealthy experience on predicting unseen categories in the presence of large domain discrepancy, which will well generalize to various openness. More specifically, Fusion Information Guided Feature Prototype Generation module is proposed to synthesize visual feature prototype conditioned on category semantic prototype in training stage. Further, Class-Aware Feature Prototype Alignment module is designed in refining stage to align the global feature prototype for each class between two domains. Experimental results verify that the proposed model not only has superiority on classifying the image instances of known and unknown classes, but also well adapts to various openness.

© 2022 Elsevier Ltd. All rights reserved.

1. Introduction

Currently, deep learning-based models heavily rely on massive labeled data. However, data annotation is an expensive and error-prone operation, which limits the practical applications of deep model to a certain extent. In addition, the model trained on massive annotated data has poor generalization ability and cannot be widely used in various new environments. In fact, when facing such open scene, humans can quickly learn a new concept from descriptive knowledge or knowledge in other fields. Inspired by this learning mechanism, an alternative is to perfectly transfer the model trained on samples of relevant fields with abundant annotation to the field of interest that lacks label information.

^{*} This work is jointly supported by the National Key Research and Development Project under Grant 2020YFB2103902, the National Science Fund for Distinguished Young Scholars under Grant 61825603, the National Natural Science Foundation of China under Grant 62001397, the Natural Science Basic Research Program of Shaanxi (Program No. 2020JQ-212), and the Fundamental Research Funds for the Central Universities under Grant 31020200QD036.

^{*} Corresponding author.

E-mail address: jiangzhiyu@nwpu.edu.cn (Z. Jiang).

Performing intelligent recognition in such open scene is quite difficult, since the practical situation in the wild is extremely complicated. On the one hand, the data collected in the real scene may deviate from the training data in terms of illumination conditions, shooting views, sensors, etc., resulting in large domain discrepancy in data distributions [1]. On the other hand, the nature is so diverse that training samples may not cover the whole species in the real environment, leading to enormous category gap in label space [2]. In summary, there exists large discrepancy in both label space and data distribution between the training samples and the testing samples. To address this issue, several methods, such as Unsupervised Open Domain Transfer Network (UODTN) [3], Open Set Domain Recognition (OSDR) [4] and Batch Nuclear-norm Maximization (BNM) [5] have been proposed. However, these methods directly fix the category relationship between the source and target domains, which is restricted in real scenarios.

In order to achieve above mentioned objectives, a generalized setting termed Adaptive Open Domain Recognition (AODR) is firstly defined in this paper. For AODR, the domain of interest which owns various novel categories termed as the target domain, and the relevant domain with sufficient supervision termed as the source domain. As samples from different domains are of-

Table 1
The major differences among CDA, OSDA, OSR, ZSL, OSDR and AODR tasks.

	Domain Discrepancy	Unknown Classes in Target Domain	Specific Labels for Unknown Classes	Proportion of Unknown Classes
CDA	✓	×	×	×
OSDA	✓	✓	×	×
OSR	×	✓	×	×
ZSL	×	✓	✓	×
OSDR	✓	✓	✓	×
AODR	✓	✓	✓	✓

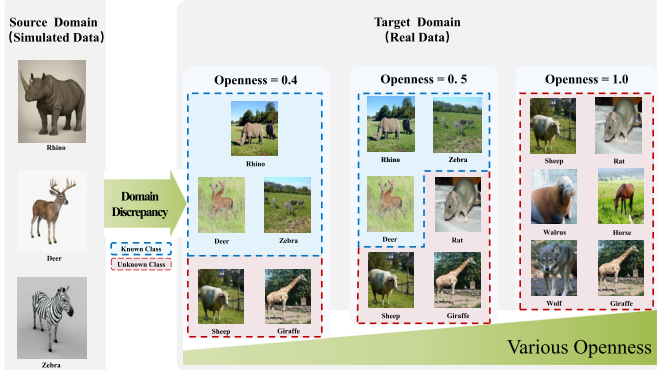


Fig. 1. Illustration of the Adaptive Open Domain Recognition (AODR). The target samples are obviously drawn from natural scene while the source samples are simulative images. In addition, the target domain contains unknown quantity of novel categories which are absent in the source domain. The task is to assign a specific label for each target sample.

ten drawn from diverse distributions, it is obvious that there exists large discrepancy in both label space and data distribution between the source and the target domain. In addition, the openness defined as the proportion of unknown categories represents the discrepancy of label sets between two domains. As shown in Fig. 1, the word “adaptive” indicates that AODR can adapt to various openness. The goal of AODR is to adaptively assign a specific classification of each sample, whether it belongs to known or unknown categories. Table 1 summarizes the main differences between AODR and other related tasks. Traditional Closed Set Domain Adaptation(CDA) [6,50] strictly assumes that the source and target domain share the same categories. Open Set Domain Adaptation(OSDA) [7] usually assumes that the source label set is a subset of the target label set. However, the OSDA task only classifies samples belonging to known classes in the target domain, while samples of unknown classes are ignored. More similar to OSDA, Open Set Recognition (OSR) [8] also classifies all unknown samples into a unified “unknown” category. The difference is that the training and testing data come from the same dataset in the OSR task and there are no domain discrepancy. Similar as OSR, the samples in Zero-Shot Learning (ZSL) [2] are also distributed in the same domain without domain discrepancy, except that ZSL identifies each sample with the help of auxiliary information. OSDR [3] attempts to assign a specific category label to each sample in the absence of any training samples for the unknown class and the presence of domain discrepancy between the source and target domain. AODR is based on OSDR and further considers the effect of openness on model performance.

AODR confronts three major technical issues. First, since nothing knowledge is known about the target label space, it is impossible to determine which parts of the source domain and the target domain should perform domain-invariant feature learning. If the model naively aligns the whole source domain with the target domain, the alignment of different label set will cause serious nega-

tive migration. Furthermore, different number of unknown classes in the target domain affects the identification results. To be more specific, the larger number of novel categories are injected into the target domain, the less sharing information can be exploited during knowledge transfer. This circumstance makes it more difficult to identify unknown classes. To solve this problem coarsely, Progressive Adaptation Network (PAN) is proposed in this work. Inspired by the effectiveness of meta-learning in several tasks [9,10], the PAN is comprised by a collection of episodes and each episode mainly consists of two stages. In this framework, each episode mimics an AODR task since different number of unknown categories randomly appear in each episode. As the episode goes on, the basic model has progressively accumulated abundant experience in intelligent recognition and feature alignment under various openness.

Second, confronted with various openness, it is rather difficult to introduce different number of intermediate categories to construct full path when adopts explicit knowledge based on knowledge graphs [3,11] as auxiliary information. A more appropriate alternative is to utilize implicit knowledge representations, such as word embeddings [12] and category attributes [13]. Furthermore, category attributes has a detailed visual feature description, which can describe a category more clearly and intuitively. However, the subjective deviation and limited dimension also greatly limit its development. Instead, the word embeddings are generated by learning the characteristics of category words in a large corpus through machine learning algorithm. These generated vectors have extremely rich semantic characteristics. However, there is a semantic deviation between the category expressed by the word vectors and the category expressed by the visual features, making great deviation to the experimental results. To solve this problem, Fusion Information Guided Feature Prototype Generation (FGPG) module is studied to fuse category attributes and word embeddings, and then construct the interaction between visual features and category semantics through generative adversarial networks.

Third, the number of novel category in the target domain make a significant impact on the effectiveness of domain adaptation. If directly align the entire distribution under various openness, such as Maximum Mean Discrepancies (MMD) [14], Joint Maximum Mean Discrepancies (JMMD) [15] and Deep CORrelation ALIGNment (DCORAL) [16], it will lead to the whole samples from two domains are aligned, but not between categories. Motivated by the prototype-based approach in few-shot recognition [2,17] and cross-domain tasks [18], Class-Aware Feature Prototype Alignment (CFPA) module is proposed to achieve feature adaptation in category level.

In summary, the main contributions of this paper are listed as follows:

- (1) For open domain recognition task, the known and unknown category overlap defined as openness can affect the recognition model a lot. In this work, a more realistic setting termed Adaptive Open Domain Recognition is firstly introduced to consider this issue. Furthermore, to achieve adaptive transfer, Progressive Adaptation Network is designed to progressively accumulate wealthy experience on predicting unseen categories in the

presence of large domain discrepancy. It can well generalize to various openness.

- (2) Both word vector and category attribute can be regarded as representation of object in different aspect. However, the semantic deviation exist. Therefore, Fusion Information Guided Feature Prototype Generation is proposed to obtain more abundant and accurate descriptive information by fusing them in training stage. FGPG is more robust to various openness since it directly constructs the interaction between visual features and fused category semantics through generative adversarial networks.
- (3) Aligning the entire feature distribution under various openness is essential and misalignments of samples belonging to the same class between two domains are always ignored due to the entire alignment. Therefore, Class-Aware Feature Prototype Alignment is proposed to align the global feature prototype between two domains in refining stage. The global feature prototype is adaptively updated in each episode to suppress the negative effects of false pseudo-label and misaligned categories, and further adapt to various openness.

The remaining parts of this article are arranged as follows. In Section 2, mainstream unknown classes recognition methods and several unsupervised domain adaptation methods are reviewed. Then, a detailed description of the proposed method is introduced in Section 3, and the experiments are shown in Section 4. Finally, the conclusion is presented in Section 5.

2. Related works

In this section, previous works related to the proposed method are reviewed.

2.1. Unknown classes recognition

With the rapid advancement of deep model [19,20], most recent approaches achieving unknown categories recognition can be mainly mapped into two research streams. The first popular approach adopts knowledge graph to transfer knowledge. In this approach, knowledge is explicitly represented as rules or relationships among objects that can be adopted to learn visual classifiers for unknown categories. Given classifiers of original visual concept as inputs, Misra et al. [21] apply a straightforward combination regulation to generate a classifier for novel complex concept. Inspired by the fact that an object can not be dog and cat simultaneously, Deng et al. [22] use this exclusion rule as a constraint on the loss during training object classifiers. Different from the previous methods that use graph as constraints, methods proposed in [11] directly utilize the knowledge graph to generate unknown object classifiers. One obvious shortcoming of these methods is that they are sensitive to openness, because the proportion of unknown classes heavily affects the available known information used to learn class relationships.

The second popular approach adopts semantic representations. The main idea exploits learned semantic vectors from text data to represent each class and then directly learns the mapping relationship between semantic vectors and visual features. Early works on classifying unknown classes utilize category attributes representing each class since the category attributes contain abundant visual features. Jiang et al. [23] proposed to exploit dictionary learning framework to directly model the latent attribute space, where the images can be reconstructed by some latent attribute dictionary items. In addition, SAE [24] is proposed to preserve all information contained in the original visual features by additional reconstruction constraint. Recently, generative methods [25,26] have shown great potential in unknown classes recognition by synthesizing features of unseen classes based on attribute descriptions,

which transform the problem into the traditional supervised learning with full training samples. However, in the process of annotating attributes, subjective bias is prone to occur. Therefore, Rohrbach et al. [27] excavate these associations from diverse internet sources to replace manually designed attributes. In [28], two different convolutional neural networks are trained in an unsupervised way obtaining language knowledge and image feature, and then a linear mapping is learned between the word embeddings and feature representations. Inspired by this work, DeVise [29] is put forward to learn a mapping from visual features to word embeddings by using a transformation layer and a ConvNet layer. These methods based on word embeddings, which do not need manual annotation but is quickly generated by machine learning algorithm from a large number of corpus, have achieved great success on unknown categories recognition. However, the semantics of category represented by the word embeddings based only on category words, and the semantic representation of category represented by visual features are biased. In order to reduce the semantic interval, it is necessary to make full use of more abundant and accurate information to correctly represent categories.

2.2. Unsupervised domain adaptation

Domain adaptation [30], which aims to use knowledge learned from the source domain to boost the performance of the target domain, has been extensively researched in numerous visual tasks. With the rapid development of convolutional neural networks, learning domain-invariant features is widely utilized to reduce domain shift [31]. Approaches along this main line may be broadly divided into metric criterion-based method and adversarial learning-based method.

The metric criterion-based method adapts the domain distribution by minimizing the statistical distance between deep features. For example, to minimize Maximum Mean Discrepancy (MMD) [51], only one single linear kernel is applied to the fully connected layer in DDC [32]. Deep Adaption Network (DAN) [14] considers the sum of MMD defined among several fully connected layers. In the Joint Adaptation Network [15], instead of independent alignment on marginal or conditional distribution, which usually requires strong assumption of independence and smoothness for the factorized distribution, joint distribution differences of multi-layer activation are considered. PMD [33] attempts to minimize the first-order Wasserstein distance of minimum weight graph matchings between two domains. In addition, domain discrepancy can be also measured on the differences between the second-order statistic such as covariance [16,34]. Further, Xu et al. [35] proposed a simple yet effective statistic distance to characterize the mean-feature-norm discrepancy across domains to address the problem that existing statistic divergences may fail to precisely depict the domain shift.

For adversarial learning-based method, the minimax optimization is performed between the introduced domain discriminator and the visual feature extractor. In [36,37], Adversarial Dropout Regularization (ADR) and Maximum Classifier Discrepancy (MCD) are proposed to train the deep model in an adversarial way to generate discriminative features located in the vicinity of the decision boundary. On the basis of the original source-target discriminator, the local subdomain discriminator is added in [38]. In this method, each class must be determined whether it comes from the source dataset or the target dataset, and then a dynamic adversarial factor is designed to dynamically adjust the loss of both proportion. Multiple discriminators may strengthen the discriminating ability for better collaboration with the generator in some situations, while the overly multiple discriminators may also break the balance of adversarial training. To address this issue, Cui et al. [39] proposed

to pursue a more balanced minmax adversarial game by building the bridge on both generator and discriminator.

Although a great deal of success has been achieved, most of them only enable to handle relatively simple tasks. As for AODR, it is rather difficult to determine appropriate metrics due to the uncertain number of unknown classes. In addition, using the generative adversarial network to align the overall distribution from the domain level also results in misalignment between the same categories belonging to two domains.

3. Proposed method

In this section, the problem formulation is introduced firstly and then proposed approaches are described in detail.

3.1. Problem formulation

Suppose that there exist a source domain $S = \{x_i, y_i, a_i, w_i | 1 \leq i \leq N_s\}$ which consists of N_s training samples and a target domain $T = \{x_i, a_i, w_i | 1 \leq i \leq N_t\}$ which consists of N_t unlabeled testing samples, where $x_i \in R^D$ denotes the D -dimensional visual feature representations for the i th object instance, y_i is its class label, $a_i \in R^K$ and $w_i \in R^W$ represent its K -dimensional category attributes and W -dimensional word embeddings respectively. It's worth noting that the source samples are drawn from distribution p but the target samples from distribution q , and $p \neq q$. Obviously, domain discrepancy exists in the source domain and the target domain from both data distributions and label space. We use C_s and C_t to denote the label set of source domain and the target domain, respectively. Therefore, the openness between two domains can be calculated as $1 - \frac{|C_s \cap C_t|}{|C_s \cup C_t|}$. The goal for Adaptive Open Domain Recognition (AODR) is to adaptively assign a specific label for each target sample with the help of auxiliary information, whether it belongs to known or unknown categories.

3.2. Two-Stage progressive adaptation network

To achieve an adaptive transfer model which aims to reduce the interference of openness on the process of unknown class recognition and domain alignment, the Progressive Adaptation Network (PAN) framework is introduced. As exhibited in Fig. 2(a), PAN consists of multiple episodes and each episode is composed of a model training and a model refining stage. During each episode, several labeled source image instances are stochastically chosen as the training set, and several unlabeled target instances are randomly taken as the refining set. In the training stage, samples from training set are embedded into the Fusion Information Guided Feature Prototype Generation (FGPG) module, which is applied to predict class labels from the corresponding category semantic prototype. In the refining stage, samples from refining set are firstly embedded into the FGPG module to generate pseudo-label. Then the Class-Aware Feature Prototype Alignment (CFPA) module is utilized to measure domain discrepancy between the source domain and the target domain by the differences among global prototypes for each class. The entire module is updated by back-propagating the gradients of the total loss which generated from the training stage and the refining stage. The final loss can be formulated as:

$$\mathcal{L} = \mathcal{L}_{\text{train}} + \mathcal{L}_{\text{refine}}. \quad (1)$$

Updating each episode can be regarded as a procedure of accumulating experience on Adaptive Open Domain Recognition (AODR). As the episode progresses, these experience will be delivered to the next episode. After multiple episodes, the model can be expected to become an expert at predicting all samples belonging to both known and unknown classes in the presence of large domain discrepancy. In this circumstance, the final obtained model

can well generalize to various openness. The training module and the refining module in one episode will be described detailedly in the following section.

3.3. Fusion information guided feature prototype generation

The key to achieve identification of unknown classes lies in how to construct the relationship between known and unknown classes for carrying out knowledge transfer. This is usually transformed into how to describe a class. In order to obtain more abundant and accurate semantic description information to correctly represent categories, Fusion Information Guided Feature Prototype Generation (FGPG) is designed in the training stage. The FGPG makes full use of existing auxiliary information to extract advantages and meanwhile avoid shortcomings of existing descriptive method. First of all, FGPG integrates the word embeddings which are extracted by the Glove Text model [40] into category attributes to enrich the descriptive information. Specifically, word vectors are embedded into the category attributes space by a continuously learning mapping network $M: R^W \rightarrow R^K$, and then fuse the two in accordance with a certain proportion coefficient, which can be formulated as:

$$f_i = \rho M(w_i) + (1 - \rho)a_i \quad (2)$$

where $w_i \in R^W$ and $a_i \in R^K$ denotes word embedding and category attribute of the i th sample respectively. The ρ means fusion coefficient.

After obtaining rich fusion information, we regard it as category semantics. FGPG firstly synthesizes visual prototype conditioned on the descriptive category semantic prototype in a generative adversarial manner. Specifically, a class semantic generating network $C: R^D \rightarrow R^K$ is learned to perform the visual features into the category semantic space. This process can be achieved by regressing the visual features to be as close as possible with the corresponding category semantic prototype. Similarly, a visual prototype generating network $G: R^K \rightarrow R^D$ is learned to project the category semantic prototype into the image feature space. Notice that every category always contains plenty of image objects but all of them corresponding to only one category semantic prototype. Under this circumstance, the mapping network G just learns to generate visual prototype at the category level from corresponding category semantic prototype. Further, this is accomplished by minimizing the distance between the generated visual feature prototype and the authentic visual feature. Therefore, the entire generation loss can be formulated as:

$$\mathcal{L}_{\text{rec}} = \sum_i \|C(x_i) - f_i\|_2^2 + \sum_i \|G(f_i) - x_i\|_2^2 \quad (3)$$

where $f_i \in R^K$ denotes the real fusion information of the i th sample. The $x_i \in R^D$ represents feature representation of the i th sample. Further, $C(x_i) \in R^K$ denotes the generated class semantics and $G(f_i) \in R^D$ means the generated feature prototype.

With the class semantic generating network C and the visual prototype generating network G , the interactions between the visual feature space and the category semantic space can be constructed. Nevertheless, they are separate to each other. In order to better maintain semantic consistency, a discriminator D is introduced to strengthen the interaction between the two mapping networks through adversarial mechanism, as shown in Fig. 2(b). The loss is written as:

$$\mathcal{L}_{\text{gan}} = \min_{G, C} \max_D E[D(x, f)] - E[D(C(x), G(f))]. \quad (4)$$

Finally, the multi-modal classification loss is utilized to exploit discriminative representations for distinguishing categories. This loss function interweaves class labels, image features, and category semantics into a unified framework, which is profitable for

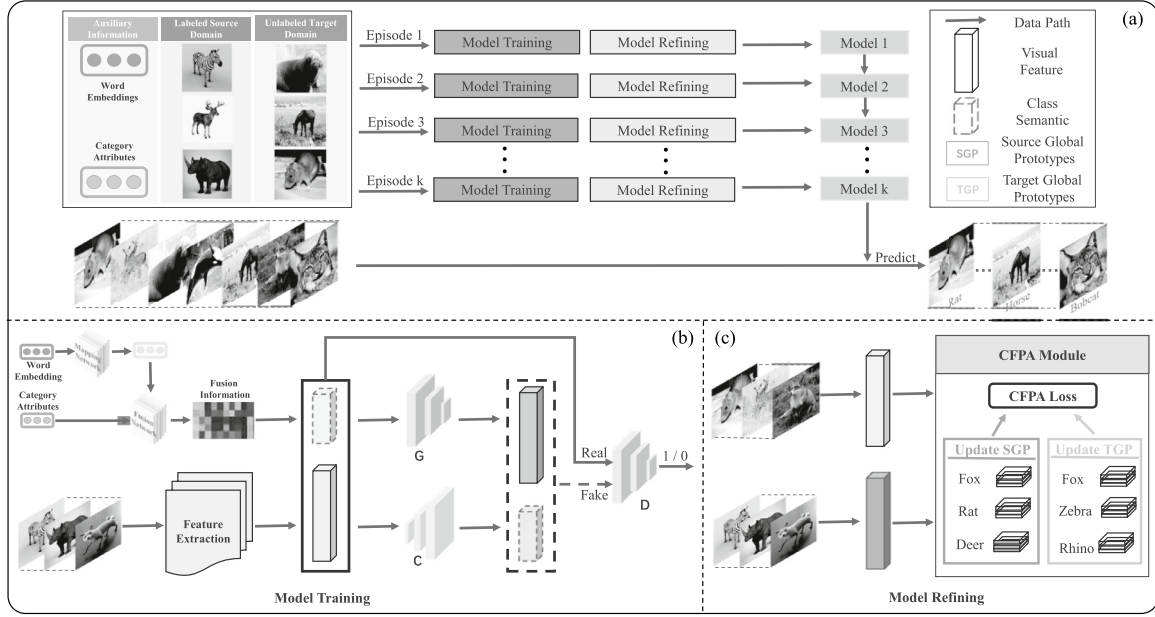


Fig. 2. (a) Overview of the proposed Progressive Adaptation Network. Given labeled source samples, unlabeled target samples and category semantics as input, the model enable to progressively accumulate wealthy experiences on predicting unseen categories in the presence of large domain discrepancy. Note that the optimization model after updating by the k th episode termed as model k and the label of target samples are predicted by the final model. (b) Details of Fusion Information Guided Feature Prototype Generation module during training stage, which can obtain more abundant and accurate descriptive information by fusing category attributes and word embeddings, and then synthesize visual prototype conditioned on the descriptive category semantic prototype. The G, C and D denote the visual prototype generating network, class semantic generating network and discriminator, respectively. (c) Details of Class-Aware Feature Prototype Alignment module during refining stage, aligns the global feature prototypes between the source domain and the target domain.

the final category prediction. Specially, with the above mentioned model, the category semantics prototype of each training sample is embedded into the visual feature space to acquire its corresponding category visual feature prototype. And then these obtained feature prototypes are stored in a feature buffer. Further, the affinities between each visual object and all category visual prototypes can be acquired by their inner product. Similarly, in the category semantic space, the affinities between generated category semantic prototype of each visual object and all category semantics can be acquired by their inner product. In this case, the classification probability of each input image instance x that belongs to the i th class can be evaluated as:

$$\mathcal{L}_{cla} = -\sum_x \log \frac{\exp(x^T G(f_i))}{\sum_j \exp(x^T G(f_j))} - \sum_x \log \frac{\exp(C(x)^T f_i)}{\sum_j \exp(C(x)^T f_j)} \quad (5)$$

where $G(f_i) \in \mathbb{R}^D$ denotes the visual feature prototype of the i th category and $C(x) \in \mathbb{R}^K$ signifies the class semantic prototype of the i th category. By minimizing $\mathcal{L}_{cla}$, instances belonging to the same class are enforced to possess higher similarity with their corresponding category semantic prototype. In this circumstance, we can effectively preserve discriminative information both in the visual features space and category semantics space. Finally, the full objective loss in training stage then becomes

$$\mathcal{L}_{train} = \alpha \mathcal{L}_{rec} + \beta \mathcal{L}_{gan} + \gamma \mathcal{L}_{cla} \quad (6)$$

where the α , β , γ denote the weight of each loss.

3.4. Class-Aware feature prototype alignment

Most of the existing approaches have resolved unsupervised domain adaptation via minimizing the domain differences which measured by MMD [32] between the source and the target samples, or maximizing the domain confusion by introduced domain discriminator. Regardless of the specific category of each sample, domain differences and domain confusion are measured on the entire source and target data. However, for AODR problem, if directly

align the entire distribution will lead to the whole samples from two domains are aligned, but samples belonging to the same class from two domains may be misaligned since there exists ambiguous quantity of unknown categories in the target domain. To achieve domain alignment from class level, a straightforward approach is to train a collection of domain classifiers for each class. Nevertheless, training a series of specific classifiers is inefficient. In addition, the performance of domain adaptation will be hurt by the false pseudo-label which occurred in the target domain.

Motivated by the prototype-based approach in few-shot recognition [17] and cross-domain tasks [18,41], Class-Aware Feature Prototype Alignment (CFPA) module is proposed to achieve feature adaptation in category level. Rather than naively training domain classifiers for each category, CFPA attempts to preserve the semantic consistency in visual feature space via minimizing discrepancies between prototype pairs which share the same category label across domains. Formally, the prototype can be expressed as:

$$P_k^S = \frac{1}{|GT_k|} \sum_{k_i \in GT_k} F(k_i) \quad (7)$$

$$P_k^T = \frac{1}{|PT_k|} \sum_{k_i \in PT_k} F(k_i) \quad (8)$$

where P_k^S and P_k^T denote the prototype of the k th class belonging to the source and target domain respectively. $F(k_i) \in \mathbb{R}^D$ represents the feature representation of i th sample whose class label is k . GT_k denotes the set of samples with true label k in the source domain and $|GT_k|$ indicate the number of samples belonging to k th category in the source domain. Similarly, PT_k denotes the set of samples with predicted label k in the target domain and $|PT_k|$ indicate the number of samples predicted to k th category in the target domain. With the trained visual prototype generating network G, the target sample can be predicted by searching the most similar generated category visual prototype in the visual feature space with

a pre-defined distance metric. For each unknown instance x_t , its category label is acquired by,

$$\hat{y}_t = \underset{k}{\operatorname{argmin}} d(x_t, G(a_k)) \quad (9)$$

where a_k denotes the category semantics of the k th category, and $G(a_k)$ represents the corresponding generated category visual feature prototype. $d(\cdot, \cdot)$ signifies a metric function, such as the Cosine or Euclidean distance.

Note that the prototypes mentioned in Eqs. (7) and (8) are calculated over entire samples. However, in the especial process proposed in PAN, the capacity of each episode is generally small, and the image instances of source and target domain in the same episode may belong to inconsistent classes, which makes it impractical to perform categorical adaptation for the whole categories at this episode. For example, two images are chose respectively from two domains for refining, whereas *cat* only occurs in the source object. As a consequence, the prototype of *cat* can not be aligned across domains at this episode. To tackle this issue, we dynamically obtain global feature prototypes, and adaptively update them by local prototypes at each episode, which can be formulated as follows:

$$\delta = \operatorname{sim}(P_k^{(i)}, GP_k^{(i-1)}) = \left(\frac{(P_k^{(i)})^T GP_k^{(i-1)}}{\|P_k^{(i)}\| \|GP_k^{(i-1)}\|} + 1 \right) / 2 \quad (10)$$

$$GP_k^i = \delta P_k^{(i)} + (1 - \delta) GP_k^{(i-1)} \quad (11)$$

where $\operatorname{sim}(\cdot, \cdot)$ represents the cosine similarity. $P_k^{(i)}$ is the local prototype of the k th class at i th episode. $GP_k^{(i)}$ is the global prototype of the k th class at i th episode. $\|P_k^{(i)}\|$ denotes 2-norm of the local prototype vector and $\|GP_k^{(i-1)}\|$ denotes 2-norm of the global prototype vector. The initial global prototype $GP_k^{(0)}$ is calculated by Eq. (7) for the source domain and Eq. (8) for the target domain.

Domain alignment is achieved by minimizing discrepancies between the global prototypes of the common categories shared by the source domain and the target domain. The objective function of the CFPA module can be expressed as following:

$$\mathcal{L}_{\text{refine}} = \sum_k \|GP_k^{S(i)} - GP_k^{T(i)}\|_2^2 \quad (12)$$

where $GP_k^{S(i)}$ and $GP_k^{T(i)}$ represent the global prototype of the k th source and target class at i th episode respectively.

4. Experiments

In order to elaborate the superiority of the proposed prototype based two-stage Progressive Adaptation Network (PAN) framework, several comparative experiments are performed in this section. Four datasets and implementation details are introduced firstly, and then the experimental results and relative discussions are described.

It is worth noting that the objective of AODR is to design a model which does not know the relationship of label sets between two domains but works well across various scenes. To quantitatively demonstrate the transferability of the model, we firstly explore the performance with respect to a wide spectrum of openness value. Obviously, the larger openness is, the less sharing knowledge is and the more difficult the recognition is.

4.1. Dataset description and evaluation metrics

Four datasets are involved in this work, including three coarse-grained datasets, namely I2Awa, Syn2VOC, CALTECH2LAD and

Table 2
The Statistics of Four Datasets.

	Information		Source Domain		Target Domain	
	A	W	Images	Classes	Images	Classes
I2Awa	85	300	2970	40	37322	50
Syn2VOC	64	300	59229	9	2372	15
Caltech2LAD	359	300	6603	46	33903	100
Nabirds2CUB	312	-	15381	135	11788	200

one fine-grained dataset Nabirds2CUB. As is shown in Fig. 3, for I2Awa and Syn2VOC, the source samples are simulated data, while the target samples are realistic data. For CALTECH2LAD and Nabirds2CUB, both the source and target images are real data but drawn from distinct datasets. The details of these four datasets are listed in Table 2, where A denotes category attributes and W means word embeddings. It is worth noting that category attributes are inherent to each dataset and word embeddings of each class are extracted by the GloVe text model [40] which is trained on Wikipedia corpus. A brief introduction for four datasets is given as follows.

- 1) **I2Awa**: The source domain of I2Awa is a simulated dataset with 2970 simulated images belonging to 40 classes, which is collected by Zhuo et al. [3] via Google image searching engine. The target domain of I2Awa is Awa2 [42], which contains 50 animal classes. There are altogether 37,322 images in Awa2, and an average of 746 images per category. In this dataset, the source and target domain share 40 common categories. Each category is described by 85-dimensional class attributes and 300-dimensional word embeddings.
- 2) **Syn2VOC**: The source domain of Syn2VOC consists of 59,229 synthetic images generated by CAD models from different views and under different lighting conditions [43]. The target domain contains 15 categories from the Pascal VOC2008 [44], including a total of 2372 images. In this dataset, the source and target domain share 9 common categories. Each category is annotated with 64-dimensional class attributes and 300-dimensional word embeddings.
- 3) **CALTECH2LAD**: The source domain of CALTECH2LAD contains 46 categories from CALTECH256¹. The target domain consists of 33,903 images belonging to 100 classes from LAD [45]. In this dataset, the source and target domain share 46 common categories. Each category is described by 359-dimensional class attributes and 300-dimensional word embeddings.
- 4) **Nabirds2CUB**: Considering that the Nabirds [46] and CUB² share 135 common categories of birds. We select 135 classes with 15,381 images from Nabirds as source domain and the entire CUB with 11,788 images of 200 classes as target domain. Each category is only described by 312-dimensional category attributes since fine-grained dataset may have bias in retrieving word embeddings.

Following the previous open domain recognition work [3,4], classification task has been performed on the entire target domain. To demonstrate the effectiveness of proposed approach on the identification of known and unknown classes more intuitively, the classification accuracy of known classes and unknown classes are set as evaluation metrics. Assuming that the target domain contains n_k known samples and n_u unknown samples, the recognition accuracy of the known class and the recognition accuracy of

¹ http://www.vision.caltech.edu/Image_Datasets/Caltech256.

² <http://www.vision.caltech.edu/visipedia/CUB-200-2011>.

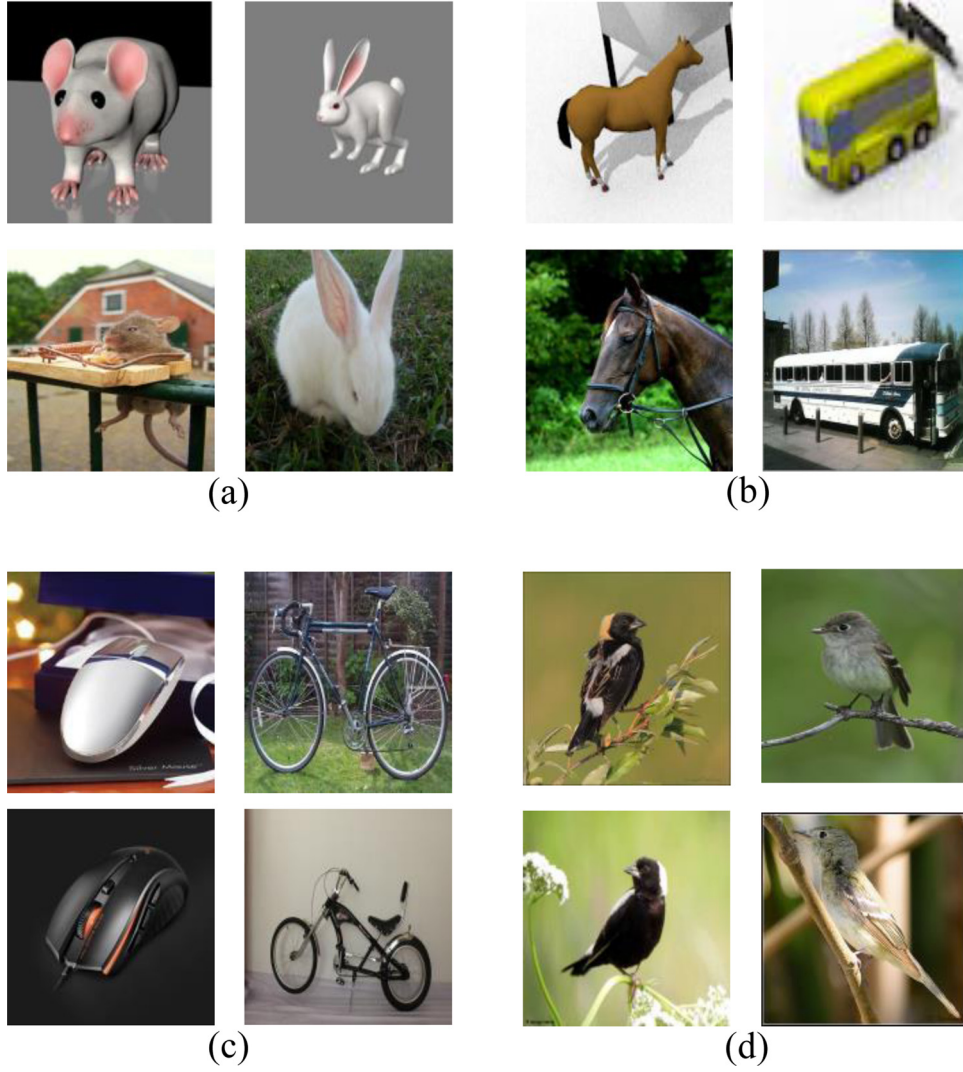


Fig. 3. The (a), (b), (c), (d) denote the data are drawn from the I2Awa, Syn2VOC, CALTECH2LAD and Nabirds2CUB dataset respectively. For each sub figure, the first row and second row show the image instances from the source domain and target domain respectively. Further, images which appear in the same column belonging to the same category.

the unknown class can be calculated as follows:

$$\text{Known} = \frac{\sum_{i=1}^{n_k} y_i = \hat{y}_i}{n_k}. \quad (13)$$

$$\text{Unknown} = \frac{\sum_{i=1}^{n_u} y_i = \hat{y}_i}{n_u}. \quad (14)$$

In addition, classification accuracy of all samples is used as an evaluation for explicitly understanding the effectiveness of the method as a whole.

$$\text{All} = \frac{\sum_{i=1}^{n_t} y_i = \hat{y}_i}{n_t}. \quad (15)$$

Considering that the trade-off between known and unknown accuracy is important, however all accuracy is greatly influenced by the number of samples and does not truly reflect the change on accuracy. If the number of known samples is large, the evaluation metric will exert too large weight on known accuracy. In this case, all accuracy mainly reflect changes on known accuracy and ignore changes on unknown accuracy. For this reason, this chapter adds H-score evaluation method. H-score is the harmonic mean of Known and Unknown, it is calculated as follows:

$$\text{H-score} = \frac{2 \times \text{Known} \times \text{Unknown}}{\text{Known} + \text{Unknown}}. \quad (16)$$

4.2. Implementation details

To extract the word embeddings of each class as one input of fusion network, we firstly utilize the GloVe text model [40] trained on Wikipedia corpus, which comes into being 300-dimensional vectors. Therefore, for each input object, its fusion category semantic can be obtained by fusing category attribute and word embedding. During training stage, the top pooling units of ResNet-50 [47] pretrained on ImageNet is used as basic model to extract image visual features. Thus, each input instance is represented as 2048-dimensional vector. The feature vector is imported into the semantic generation network to get the corresponding category description, and the fusion category semantic is imported into feature generation network to obtain the corresponding feature prototype. Further, the generated content and real content are fed into the discriminator simultaneously. The entire network is optimized in a generative-adversarial manner. During refining stage, for each input target sample, optimized ResNet-50 is utilized to obtain its feature representation. The Euclidean distance between it and the visual feature prototype generated by the optimized feature generation network can be calculated, and further the pseudo label is obtained by the nearest neighbor classification. The global feature

prototype is updated by the groundtruth of source samples and the pseudo-label of target samples. Finally, the entire network is refined by minimizing the discrepancy between the global prototypes of the common categories shared by source and target domains. In terms of the model architecture, we implemented semantic generation network, feature generation network, and discriminator network as simple 3-layer neural networks. Note that semantic generation network and discriminator use ReLU activation function while feature generation network applies the tanh function as the activation function for obtaining more stable and better results. In terms of the model parameters, the generative adversarial network and the fusion network are initialized by Kaiming initialization. Both of them are optimized with Adam optimizer whose parameters β_1 , β_2 , and ε are set to 0.5, 0.9, and 0.01, respectively. The learning rate are set to 1×10^{-5} in all experiments. Each episode usually contains 64 images, and they are randomly flipped, rotated for data enhancement, and then all of them are cropped to a size of 256×256 . The loss in the training stage is composed of three parts, the weight of them are set to 0.01, 1 and 1.5, respectively.

4.3. Comparison with state-of-the-art methods

We implemented the comparative experiments with following algorithms: zGCN [11], three improvements including UODTN [3], OSDR [4] and BNM [5], DEM [48], RN [49], CRnet [13], and E-PGN [26]. zGCN adopts word vectors extracted by the GloVe text model [40] and categorical relationship encoded in the WordNet corpus to estimate the classifiers of unknown classes. Based on zGCN, Zhuo et al. [3] design a semantic-guided matching method to align two domains and a balance constraint loss to prevent target instances of unknown classes being predicted as known classes. Further, for OSDR proposed in [4], attention-based GCN is utilized to assign distinct weight to different neighbors for learning more distinguishing feature representations. On the basis of UODTN, BNM is proposed in [5] to enforce the prediction discriminability and diversity of the unlabeled data by maximizing the nuclear-norm of the batch output matrix.

Except these four methods which use word embeddings as auxiliary information to perform unknown recognition, category attributes-based methods are also compared in our experiments. In [48], the authors proposed DEM model which utilizes the output visual feature space as the embedding space to alleviate the hubness problem. The RN is proposed in [49] to learn a general metric space for evaluating the relations between the visual features and the class semantics. Based on RN, the authors in [13] proposed CRnet to learn a more uniform visual embedding space that effectively alleviates the bias problem. Further, the E-PGN [26] model aligns the interactions between visual features and class semantics by formulating both the feature generation and the semantic inference into an adversarial network paired with a parameters-economic Cross-Entropy Loss to capture discriminative information. In the series of experiments, all hyper-parameters are fixed to the default settings in their paper.

The results on the I2AwA dataset are shown in Table 3. The best results are marked with bold font, and the second best results are marked with underline. It can be observed that zGCN can not adapt well to target samples since the classification confusion between these known and unknown instances is severe. The superiority of E-PGN lies in the multiple model Cross-Entropy loss, which preserves the discriminative knowledge and enhances the feature-semantic consistence by encoding the category semantic into the visual classification module. UODTN and OSDR display improvements on known and unknown accuracy indicating that the consideration on mitigating domain gap between the source and the target domain leads to better generalization of their model. BNM

Table 3
Classification Accuracy on The I2AwA Dataset.

	Unknown	Known	All	H-score
zGCN [11]	21.0	77.2	65.0	33.0
DEM [48]	34.7	82.8	72.6	48.9
RN [49]	28.7	83.5	71.8	42.7
CRnet [13]	38.3	77.8	69.4	51.3
E-PGN [26]	33.8	81.3	71.2	47.7
UODTN [3]	31.7	84.7	73.5	46.1
OSDR [4]	34.3	85.1	74.3	48.9
BNM [5]	<u>39.7</u>	88.3	78.0	<u>54.7</u>
PAN (Ours)	40.8	<u>87.4</u>	<u>77.6</u>	55.6

Table 4
Classification Accuracy on Three Datasets.

Dataset	Method	Unknown	Known	All	H-score
Syn2VOC	RN [49]	13.1	52.2	39.3	20.9
	CRnet [13]	8.76	68.8	48.9	15.5
	E-PGN [26]	26.5	49.8	42.1	<u>34.6</u>
	PAN (Ours)	29.5	<u>53.8</u>	<u>45.8</u>	38.1
CALTECH2LAD	RN [49]	14.1	69.5	39.4	23.4
	CRnet [13]	17.9	77.7	45.2	<u>29.1</u>
	E-PGN [26]	13.4	73.5	40.8	22.7
	PAN (Ours)	20.4	<u>77.6</u>	46.5	32.3
Nabirds2CUB	DEM [48]	19.8	53.2	42.6	28.8
	RN [49]	28.3	<u>62.1</u>	<u>51.3</u>	38.9
	CRnet [13]	36	51.6	46.6	42.4
	E-PGN [26]	22.9	57.5	46.5	32.7
	PAN (Ours)	<u>29.1</u>	62.2	51.7	<u>39.6</u>

shows remarkable improvements on known accuracy since maximize the nuclear-norm of the batch output matrix can achieve better discriminability and diversity under label insufficient learning situations. Compared with the existing state-of-the-art (SOTA), our proposed method achieves improvement on unknown accuracy by 39.7 to 40.8. This mainly due to the fusion information is utilized to learn more discriminative category representations for efficiently building the interaction between the visual and semantic prototypes. In addition, we can observe that the identification accuracy of known categories is more preferable than that of unknown categories. This result indicates the unknown samples tend to be easily misclassified as known category. It is ubiquitous that this classification shift issue exists across all existing methods. Based on the results, we can draw the conclusion that the generative model enable to assuage this shift to some extent, making the improvements of unknown accuracy.

The results on other datasets are shown in Table 4. For the Syn2VOC dataset, the entire classification accuracy is not as good as others. The reason is that there exists large openness between two domains and the category attribute is rough. As shown in Table 4, the CRnet obtains great improvement on known accuracy since the model learns a projection with high local linearity, which can cause less bias problem. However, the unknown accuracy is unsatisfactory. By contrast, the proposed model is more robust than others, which significantly improves unknown accuracy and achieves a more balanced all accuracy via boosting the unknown accuracy while preserving a high known accuracy. When comes to the CALTECH2LAD and Nabirds2CUB dataset, proposed method still shows the advantage on the unknown accuracy and all accuracy. The improvement is attributed to the two-stage progressive strategy equipped with the corresponding effective module.

4.4. Analysis on various openness

In this experiment, the impact of openness is evaluated on the I2AwA dataset. To achieve this goal, we vary the proportion of un-

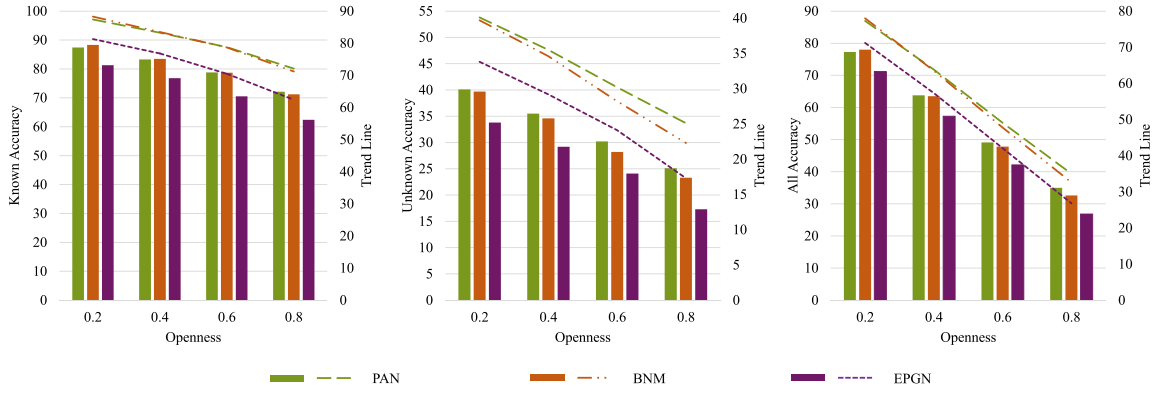


Fig. 4. The curve of classification accuracy under various openness on the I2AwA dataset. For each sub figure, the horizontal axis represents openness, and the vertical axis denotes known accuracy, unknown accuracy, and all accuracy respectively. The histogram displays the value of accuracy under various openness, and the dotted line represents the trend line for each method with the change of openness.

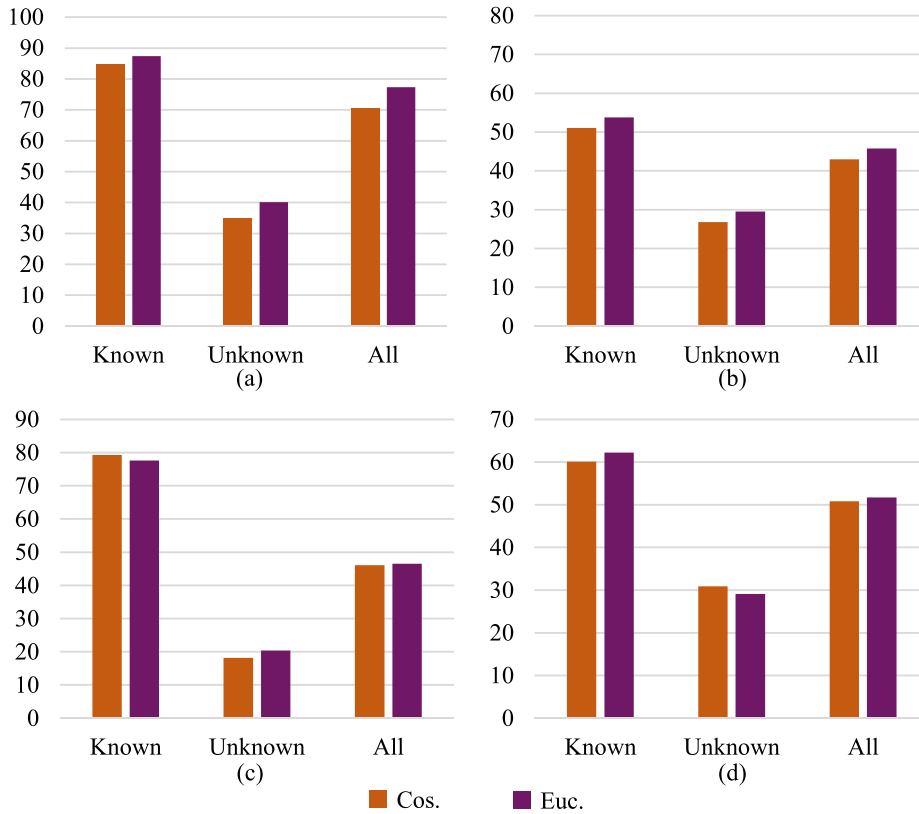


Fig. 5. Classification accuracy with Euclidean distance (short for Euc.) and Cosine distance (short for Cos.) on four datasets. Specifically, (a), (b), (c), and (d) show the results on the I2AwA, Syn2VOC, CALTECH2LAD, and Nabirds2CUB dataset, respectively.

known categories in the target domain from 0.2 to 0.8 in intervals of 0.2. We utilize histogram to show the classification accuracy under various openness, and draw a trend line for each method with the change of openness. Two representative methods are selected to perform comparative experiments. According to the exhibition in Fig. 4, we firstly observe that PAN mostly performs superior than others under various openness, which indicates that the proposed progressive two-stage training strategy can effectively adapt to diverse circumstances. Further, we can discover that the curves represent our method change slowly compared with the curves represent other methods, which means that PAN is more robust to openness change to some extent and it enable to adaptively generalize to practical situations. In addition, with the increase of open-

ness value, the curve of unknown accuracy decreased more obviously than that of known accuracy. This result reveals the fact that openness mainly affects the recognition accuracy of unknown samples, which is consistent with our original intention to make full use of precisely auxiliary information to establish visual-semantic connections for reducing the dependence on the knowledge of known samples.

4.5. Ablation study and hyperparameter study

Extensive experiments are carried out to evaluate the benefits of different components in PAN on the performance. This ablation study is conducted on the I2AwA dataset. We alternately re-

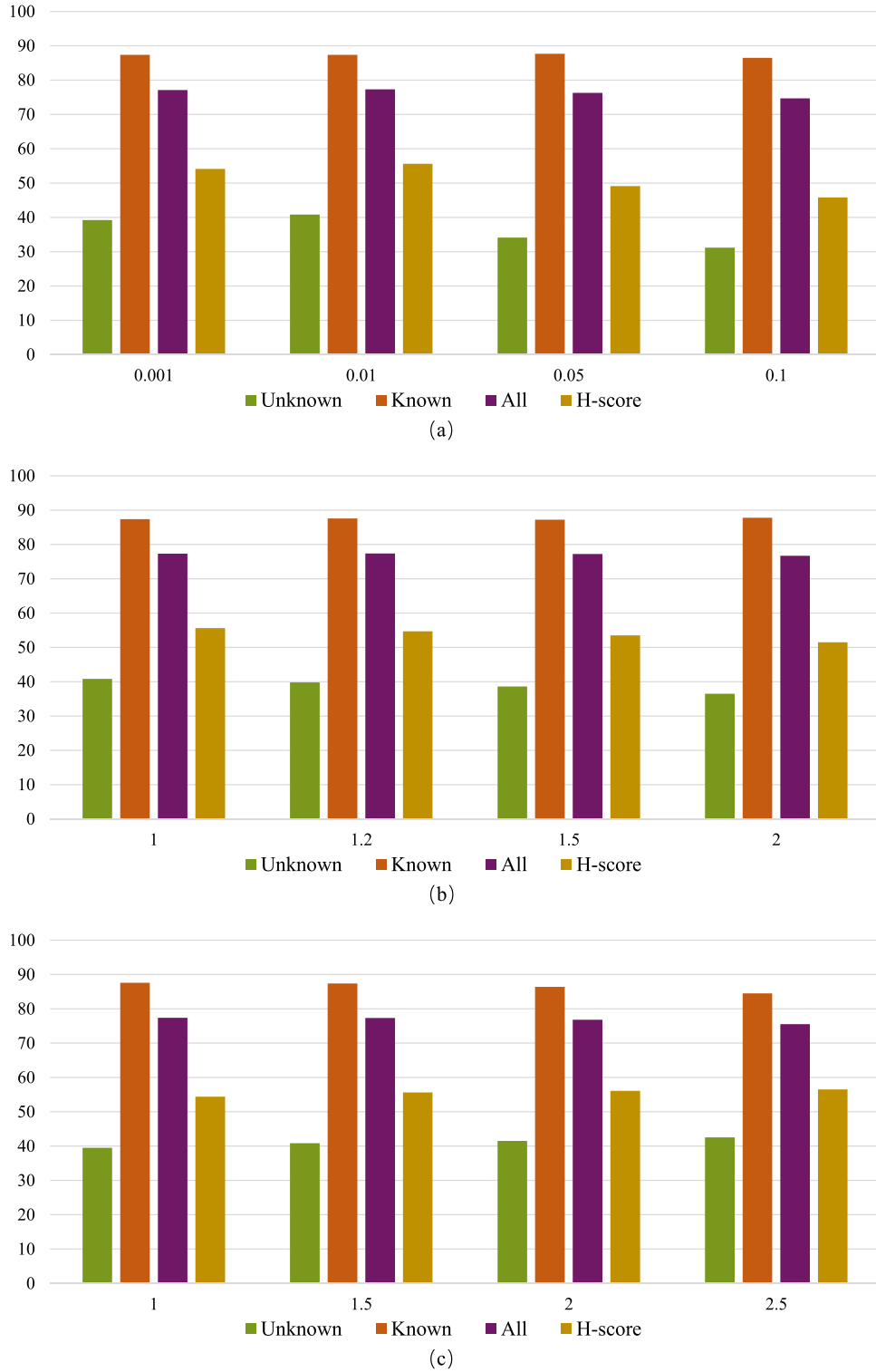


Fig. 6. Impact of different training hyperparameter settings over the performance of the proposed methods. The subplots (a)-(c) demonstrate the performance on different settings of α , β , γ in $\mathcal{L}_{\text{train}}$, correspondingly.

move the refining stage, and the information fusion module from the entire framework and report the performance. Note that removing the information fusion module means we only remove one of the word embeddings and category attributes while the other is still kept. From Table 5, we can see that the proposed method brings improvements by 6.9% on unknown categories and 4.7% on known categories with the help of class-aware feature adap-

tation during refining stage. This significant improvement validates that CFPA module can effectively avoid negative transfer and learn more domain-invariant features for effectively transferring knowledge across domains from category level. Besides, PAN outperforms word embeddings only and category attributes only by a large margin, which means the fusion strategy is efficient for open domain recognition problem. By further integrating class-aware fea-

Table 5
Ablation Study on The I2AwA Dataset.

	Unknown	Known	All
PAN (w/o refining)	33.2	82.7	72.2
PAN (w/o word embeddings)	38.7	85.6	75.6
PAN (w/o class attributes)	35.3	83.8	73.5
PAN	40.8	87.4	77.6

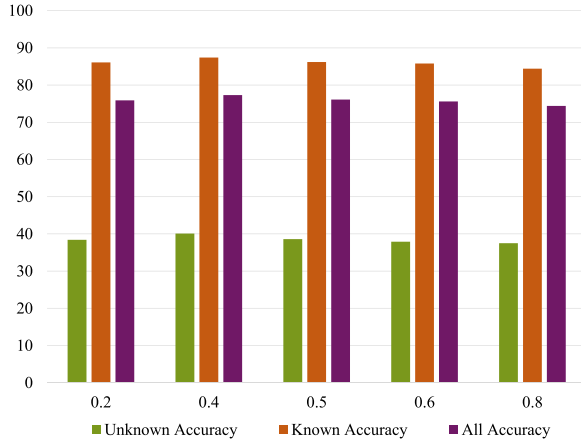


Fig. 7. The impact of different fusing hyperparameter settings over the performance of proposed methods on the I2AwA dataset.

ture alignment and fusion information guided feature generation into progressive two-stage framework, the performance gains great improvement over other method. It is reasonable as the domain discrepancy affects the knowledge transfer from known classes to unknown classes. By reducing the domain gap during refining stage, the transfer loss can be reduced to a certain extent and the classification accuracy can be improved.

Further, we quantify the impact of the hyper-parameter in fusion network on the I2AwA dataset. As shown in Fig. 7, if the fusion coefficient is too small, such as 0.2, it is difficult to excavate the advantages of word embeddings. On the contrary, if it is too large, such as 0.8, it will limit the expression of the category attributes. We can further observe that 0.4 is the optimal value for obtaining discriminative and precise descriptive information of each category.

We also study the effect of distance metric function on classification performance in this experiment. In Fig. 5, the classification accuracy under Cosine distance and Euclidean distance is compared on four distinct datasets. It can be observed that the performance obtained with Euclidean distance are substantially better than that obtained with Cosine distance under most cases. This observation implies that exploiting Euclidean distance as the metric function is more appropriate for our model.

Finally, to prove the effectiveness of different hyperparameter settings of the proposed model, we test all three hyperparameters in a range of possible values. The results are shown in Fig. 6. On the whole, the proposed model is not particularly sensitive to hyperparameter. For α , 0.01 is the optimal value. When α increases, the known accuracy remains the same, while the unknown accuracy decreases more significantly. This result indicates the generation loss mainly affects the unknown class recognition. Similar as α , the change of β mainly affects the unknown accuracy. Fig. 6(c) shows that the larger weight of classification loss, the higher known accuracy and the lower unknown accuracy. To achieve more balanced results, the optimal value of γ is 1.5.

5. Conclusion

A novel adaptive open domain recognition task has been proposed in this paper. It can be well generalized to a new scene with various openness and large domain discrepancy. Furthermore, Progressive Adaptation Network (PAN) is introduced to achieve the adaptive transfer model. The PAN is comprised by a collection of episodes and each episode mainly consists of two stages. In the training stage, fusion information guided feature prototype generation module is proposed to obtain more abundant and accurate category semantics by fusing category attributes and word embeddings, and then generate visual feature prototypes conditioned on the descriptive category semantic prototypes. In the refining stage, class-aware feature prototype alignment is utilized to alleviate negative transfer by reducing the domain discrepancy measured on the distance of global feature prototypes between the source and the target domain. Experimental results implemented on different datasets with various openness show that the proposed method is effective on classifying samples belonging to both known and unknown classes in the target domain. Ablation study results have validated the contributions of each component.

In practice, if one wants to extend a model to new scenarios, the proposed PAN can be a good candidate model. For example, when the PAN is applied to Unmanned Aerial Vehicles (UAV) systems, it only requires UAV to collect samples in a few regions rather than around the world, or to use software-generated simulated samples as training data. When UAV performs autonomous reconnaissance mission, it can accurately identify the content of the scene even when encountering a new scene. This undoubtedly will give UAV autonomous reconnaissance more economic value and great research significance. Although the PAN achieves improvement in recognition accuracy and generalization ability, the model cannot perform continuous learning which is crucial for practical application in real-world scenarios.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Q. Wang, T.P. Breckon, Cross-domain structure preserving projection for heterogeneous domain adaptation, *Pattern Recognit.* 123 (2022) 108362.
- [2] J. Zhang, Q. Li, Y.-a. Geng, W. Wang, W. Sun, C. Shi, Z. Ding, A zero-shot learning framework via cluster-prototype matching, *Pattern Recognit.* 124 (2022) 108469.
- [3] J. Zhuo, S. Wang, S. Cui, Q. Huang, Unsupervised open domain recognition by semantic discrepancy minimization, in: *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, 2019, pp. 750–759.
- [4] X. He, Y. Yuan, Z. Jiang, Open set domain recognition via attention-based GCN and semantic matching optimization, in: *Proc. International Conference on Pattern Recognition*, 2021, pp. 4626–4633.
- [5] S. Cui, S. Wang, J. Zhuo, L. Li, Q. Huang, Q. Tian, Towards discriminability and diversity: Batch nuclear-norm maximization under label insufficient situations, in: *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, 2020, pp. 3941–3950.
- [6] Q. Jiang, Y. Zhang, F. Bao, X. Zhao, C. Zhang, P. Liu, Two-step domain adaptation for underwater image enhancement, *Pattern Recognit.* 122 (2022) 108324.
- [7] R. Xu, P. Liu, Y. Zhang, F. Cai, J. Wang, S. Liang, H. Ying, J. Yin, Joint partial optimal transport for open set domain adaptation, in: *Proc. International Joint Conferences on Artificial Intelligence*, 2021, pp. 2540–2546.
- [8] L. Chambers, M.M. Gaber, DeepStreamOS: fast open-set classification for convolutional neural networks, *Pattern Recognit. Lett.* (2022).
- [9] G.J. Aguiar, E.J. Santana, A.C. de Carvalho, S.B. Junior, Using meta-learning for multi-target regression, *Inf. Sci.* 584 (2022) 665–684.
- [10] H. Xu, J. Wang, H. Li, D. Ouyang, J. Shao, Unsupervised meta-learning for few-shot learning, *Pattern Recognit.* 116 (2021) 107951.
- [11] X. Wang, Y. Ye, A. Gupta, Zero-shot recognition via semantic embeddings and knowledge graphs, in: *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 6857–6866.

- [12] X. Song, H. Zeng, S. Zhang, L. Herranz, S. Jiang, Generalized zero-shot learning with multi-source semantic embeddings for scene recognition, in: Proc. ACM International Conference on Multimedia, 2020, pp. 3976–3985.
- [13] F. Zhang, G. Shi, Co-representation network for generalized zero-shot learning, in: Proc. International Conference on Machine Learning, 2019, pp. 7434–7443.
- [14] M. Long, Y. Cao, J. Wang, M.I. Jordan, Learning transferable features with deep adaptation networks, arXiv preprint arXiv:1502.02791(2015).
- [15] M. Long, H. Zhu, J. Jia, M.I. Jordan, Deep transfer learning with joint adaptation networks, in: Proc. International Conference on Machine Learning, 2017, pp. 2208–2217.
- [16] B. Sun, K. Saenko, Deep coral: correlation alignment for deep domain adaptation, in: Proc. European Conference on Computer Vision, 2016, pp. 443–450.
- [17] Y. Chai, L. Du, J. Qiu, L. Yin, Z. Tian, Dynamic prototype network based on sample adaptation for few-shot malware detection, IEEE Trans. Knowl. Data Eng. (2022).
- [18] Y. Liu, Q. Chen, S. Albanie, Adaptive cross-modal prototypes for cross-domain visual-language retrieval, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2021, pp. 14954–14964.
- [19] Z. Jiang, Q. Wang, Y. Yuan, Modeling with prejudice: small-sample learning via adversary for semantic segmentation, IEEE Access 6 (2018) 77965–77974.
- [20] Z. Jiang, Y. Yuan, Q. Wang, Contour-aware network for semantic segmentation via adaptive depth, Neurocomputing 284 (2018) 27–35.
- [21] I. Misra, A. Gupta, M. Hebert, From red wine to red tomato: composition with context, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 1792–1801.
- [22] J. Deng, N. Ding, Y. Jia, A. Frome, K. Murphy, S. Bengio, Y. Li, H. Neven, H. Adam, Large-scale object classification using label relation graphs, in: Proc. European Conference on Computer Vision, 2014, pp. 48–64.
- [23] H. Jiang, R. Wang, S. Shan, Y. Yang, X. Chen, Learning discriminative latent attributes for zero-shot classification, in: Proc. IEEE International Conference on Computer Vision, 2017, pp. 4223–4232.
- [24] E. Kodirov, T. Xiang, S. Gong, Semantic autoencoder for zero-shot learning, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 3174–3183.
- [25] H. Kim, J. Lee, H. Byun, Discriminative deep attributes for generalized zero-shot learning, Pattern Recognit. 124 (2022) 108435.
- [26] Y. Yu, Z. Ji, J. Han, Z. Zhang, Episode-based prototype generating network for zero-shot learning, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2020, pp. 14035–14044.
- [27] M. Rohrbach, S. Ebert, B. Schiele, Transfer learning in a transductive setting, in: Proc. Advances in Neural Information Processing Systems, 2013, pp. 46–54.
- [28] R. Socher, M. Ganjoo, C.D. Manning, A. Ng, Zero-shot learning through cross-modal transfer, in: Proc. Advances in Neural Information Processing Systems, 2013, pp. 935–943.
- [29] A. Frome, G.S. Corrado, J. Shlens, S. Bengio, J. Dean, M. Ranzato, T. Mikolov, Devise: a deep visual-semantic embedding model, in: Proc. Advances in Neural Information Processing Systems, 2013, pp. 2121–2129.
- [30] C. Han, D. Zhou, Y. Xie, M. Gong, Y. Lei, J. Shi, Collaborative representation with curriculum classifier boosting for unsupervised domain adaptation, Pattern Recognit. 113 (2021) 107802.
- [31] S. Li, S. Song, G. Huang, Z. Ding, C. Wu, Domain invariant and class discriminative feature learning for visual domain adaptation, IEEE Trans. Image Process. 27 (9) (2018) 4260–4273.
- [32] E. Tzeng, J. Hoffman, N. Zhang, K. Saenko, T. Darrell, Deep domain confusion: maximizing for domain invariance, arXiv preprint arXiv:1412.3474(2014).
- [33] J. Chen, C. Li, Y. Ru, J. Zhu, Population matching discrepancy and applications in deep learning, in: Proc. Advances in Neural Information Processing Systems, 2017, pp. 6262–6272.
- [34] J. Zhuo, S. Wang, W. Zhang, Q. Huang, Deep unsupervised convolutional domain adaptation, in: Proc. ACM International Conference on Multimedia, 2017, pp. 261–269.
- [35] R. Xu, G. Li, J. Yang, L. Lin, Larger norm more transferable: an adaptive feature norm approach for unsupervised domain adaptation, in: Proc. IEEE International Conference on Computer Vision, 2019, pp. 1426–1435.
- [36] K. Saito, Y. Ushiku, T. Harada, K. Saenko, Adversarial dropout regularization, arXiv preprint arXiv:1711.01575(2017).
- [37] K. Saito, K. Watanabe, Y. Ushiku, T. Harada, Maximum classifier discrepancy for unsupervised domain adaptation, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 3723–3732.
- [38] C. Yu, J. Wang, Y. Chen, M. Huang, Transfer learning with dynamic adversarial adaptation network, in: Proc. IEEE International Conference on Data Mining, 2019, pp. 778–786.
- [39] S. Cui, S. Wang, J. Zhuo, C. Su, Q. Huang, Q. Tian, Gradually vanishing bridge for adversarial domain adaptation, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2020, pp. 12455–12464.
- [40] J. Pennington, R. Socher, C.D. Manning, Glove: global vectors for word representation, in: Proc. Conference on Empirical Methods in Natural Language Processing, 2014, pp. 1532–1543.
- [41] Y. Zheng, D. Huang, S. Liu, Y. Wang, Cross-domain object detection through coarse-to-fine feature adaptation, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2020, pp. 13766–13775.
- [42] Y. Xian, B. Schiele, Z. Akata, Zero-shot learning-the good, the bad and the ugly, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 4582–4591.
- [43] X. Peng, B. Usman, K. Saito, N. Kaushik, J. Hoffman, K. Saenko, Syn2Real: a new benchmark for synthetic-to-real visual domain adaptation, arXiv preprint arXiv:1806.09755(2018).
- [44] A. Farhadi, I. Endres, D. Hoiem, D. Forsyth, Describing objects by their attributes, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2009, pp. 1778–1785.
- [45] B. Zhao, Y. Fu, R. Liang, J. Wu, Y. Wang, Y. Wang, A large-scale attribute dataset for zero-shot learning, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition Workshops, 2019.
- [46] G. Van Horn, S. Branson, R. Farrell, S. Haber, J. Barry, P. Ipeirotis, P. Perona, S. Belongie, Building a bird recognition app and large scale dataset with citizen scientists: the fine print in fine-grained dataset collection, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2015, pp. 595–604.
- [47] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.
- [48] L. Zhang, T. Xiang, S. Gong, Learning a deep embedding model for zero-shot learning, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 2021–2030.
- [49] F. Sung, Y. Yang, L. Zhang, T. Xiang, P.H. Torr, T.M. Hospedales, Learning to compare: relation network for few-shot learning, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 1199–1208.
- [50] Z. Wang, B. Du, Y. Guo, Domain adaptation with neural embedding matching, IEEE Transactions on Neural Networks and Learning Systems 31 (7) (2019) 2387–2397.
- [51] Z. Wang, B. Du, W. Tu, L. Zhang, D. Tao, Incorporating distribution matching into uncertainty for multiple kernel active learning, IEEE Transactions on Knowledge and Data Engineering 33 (1) (2019) 128–142.



Yuan Yuan (M05CSM09) is currently a Full Professor with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, PR China. She has authored or coauthored over 150 papers, including about 100 in reputable journals, such as the IEEE TRANSACTIONS and Pattern Recognition, as well as conference papers in CVPR, BMVC, ICIP, and ICASSP. Her current research interests include visual information processing and image/video content analysis.



Xinxing He received the BE degree with the School of Computer Science and Technology from Changan University, Xi'an, China, in 2019. She is currently pursuing the MS degree with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, PR China. Her research interests include transfer learning and computer vision.



Zhiyu Jiang received the BE degree in Intelligent Science and Technology from the Xidian University, Xian, China, in 2013, and the PhD degree in Signal and Information Processing from the University of Chinese Academy of Sciences, Beijing, China, 2018. He is currently an Associate Professor with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, PR China. His current research interests include remote sensing and computer vision.